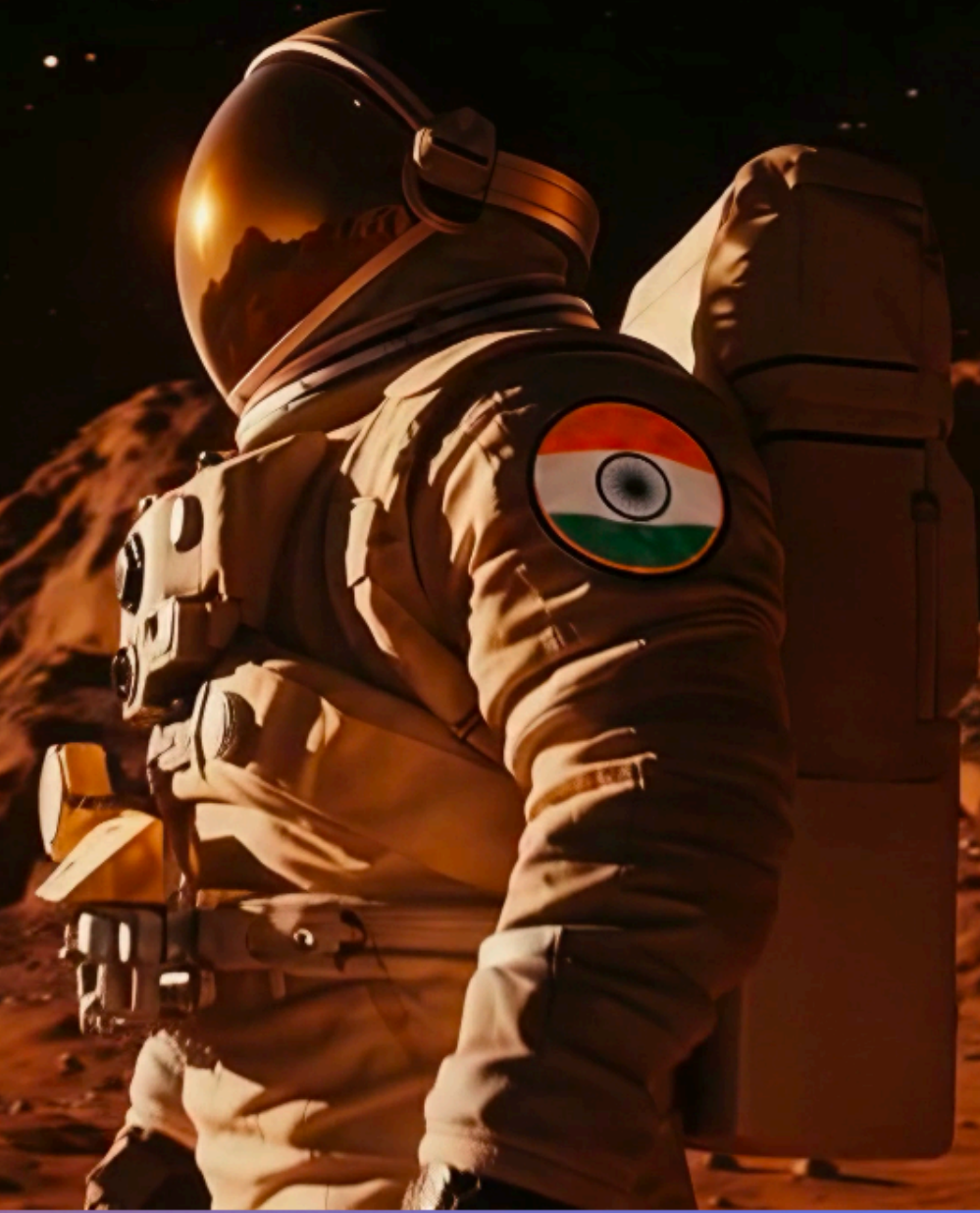




HACK2SKILL

BHARATIYA ANTARIKSH HACKATHON 2026



Team Name : **Madras Intelligence**

Problem Statement : **Route Resilience - Occlusion-Robust Road Extraction & Graph-Theoretic Criticality Analysis for Urban Mobility**

TEAM MEMBERS

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TEAM MEMBER 4

NAME : **MAHIR ALI T**

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THE PROBLEM

Satellite road extraction frequently **breaks** down under shadows, tree canopies, and urban clutter, producing fragmented road masks.

These broken networks fail when it matters most rendering the data unusable for **emergency** response, navigation, and urban planning.



Why Do We Need This Solution?

The Gap



Indian cities are expanding faster than formal road mapping



Broken road masks are unusable for disaster response or urban planning

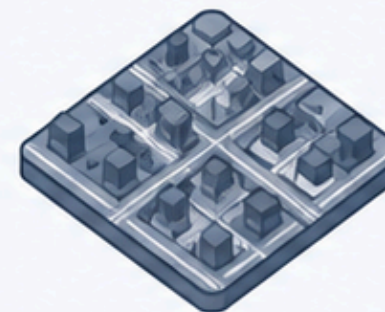


Recognized as a critical gap in national infrastructure planning

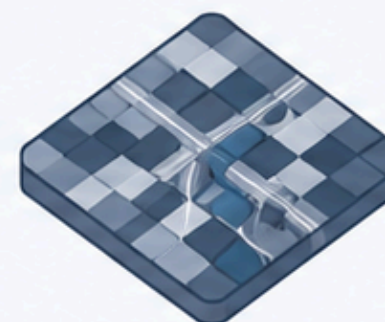
We Close This Gap



What ISRO Already Has



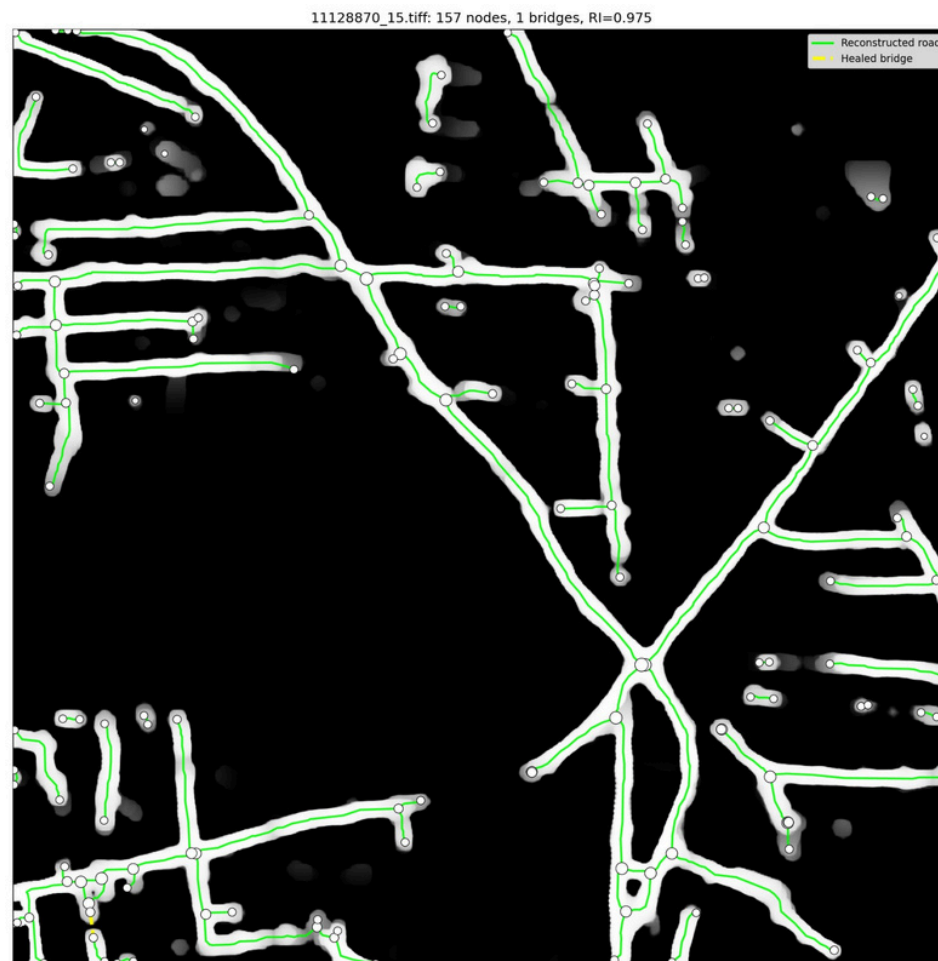
Cartosat-3, ~1m resolution



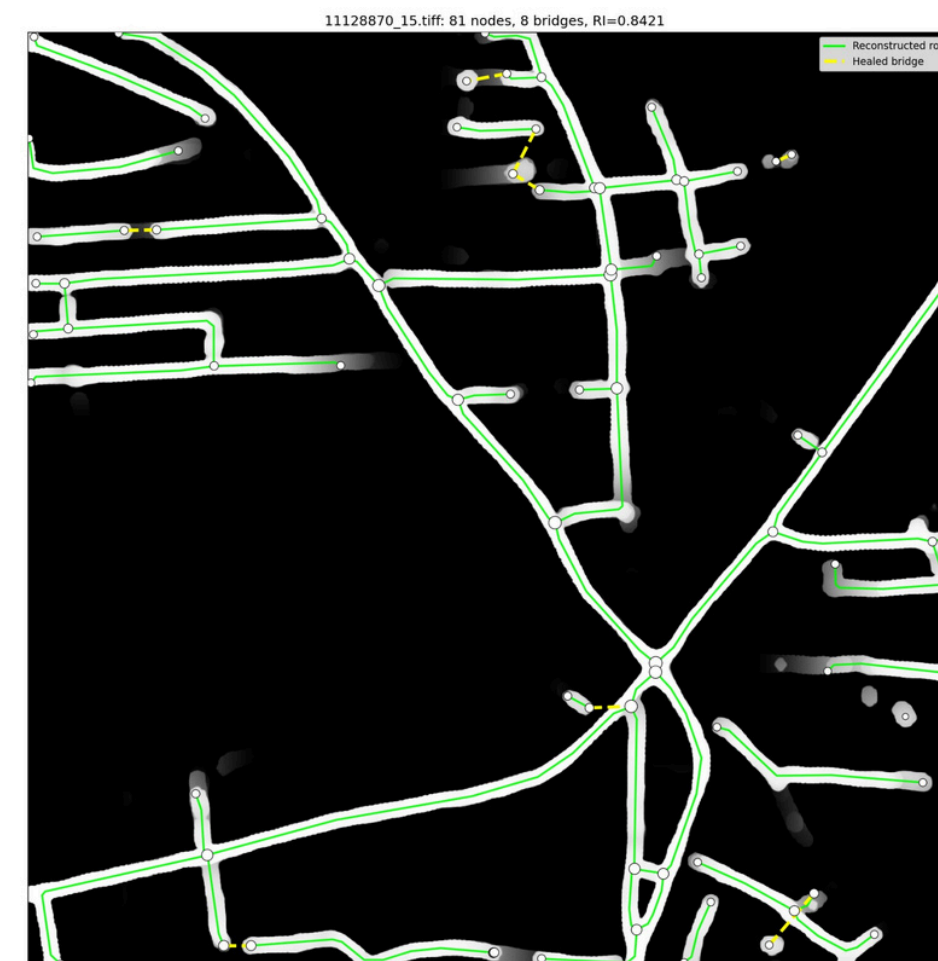
Resourcesat LISS-IV, ~5m resolution

Closing the Gap Between Capture and Action

HEALING PROCESS



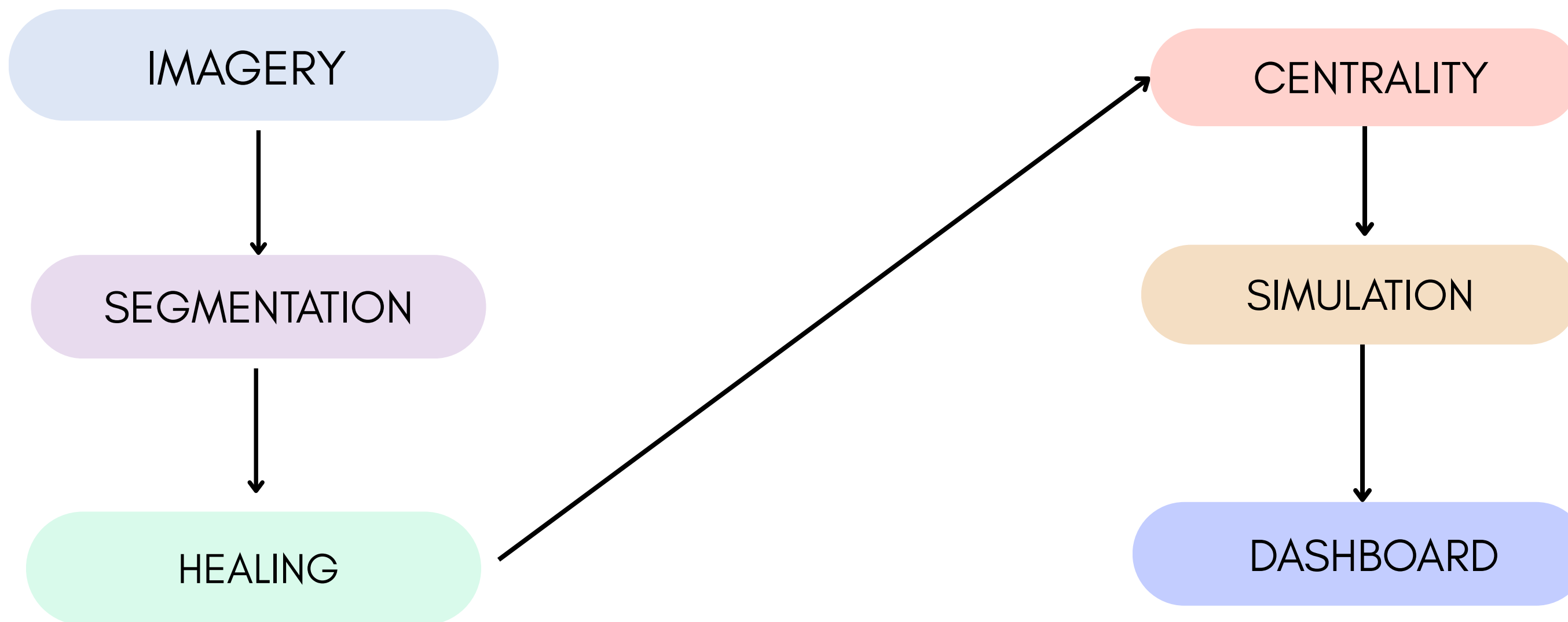
WITHOUT HEALING



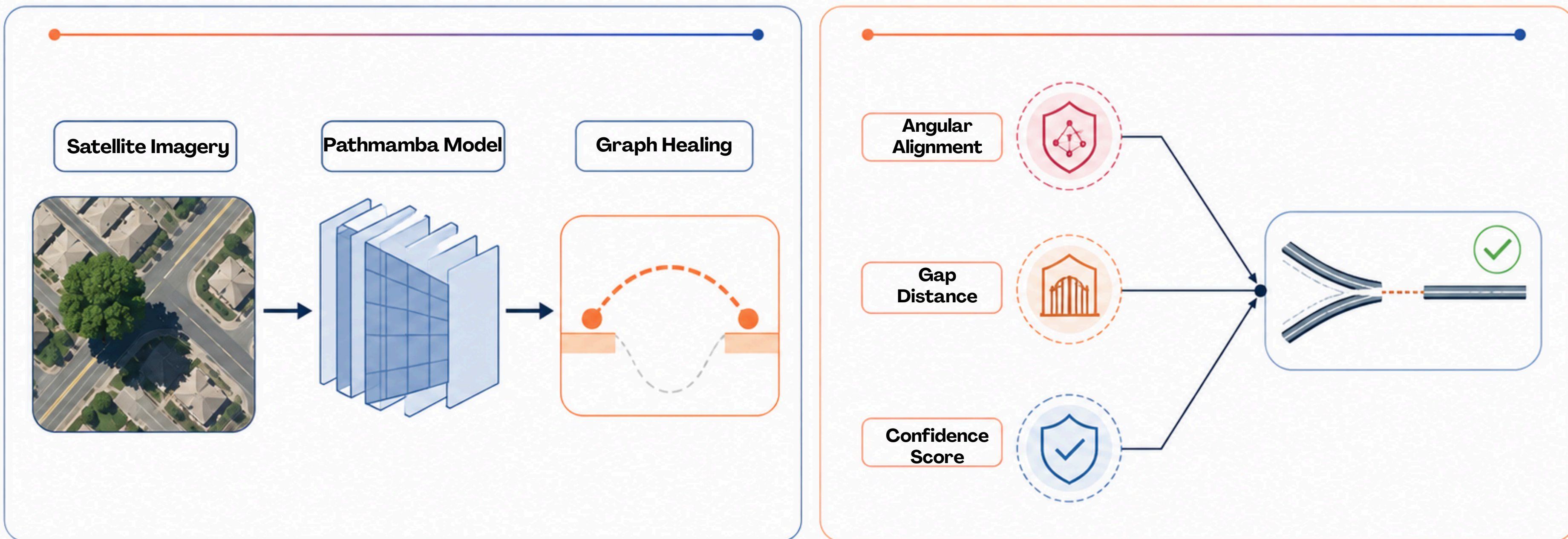
WITH HEALING

Solution overview

THE SOLUTION

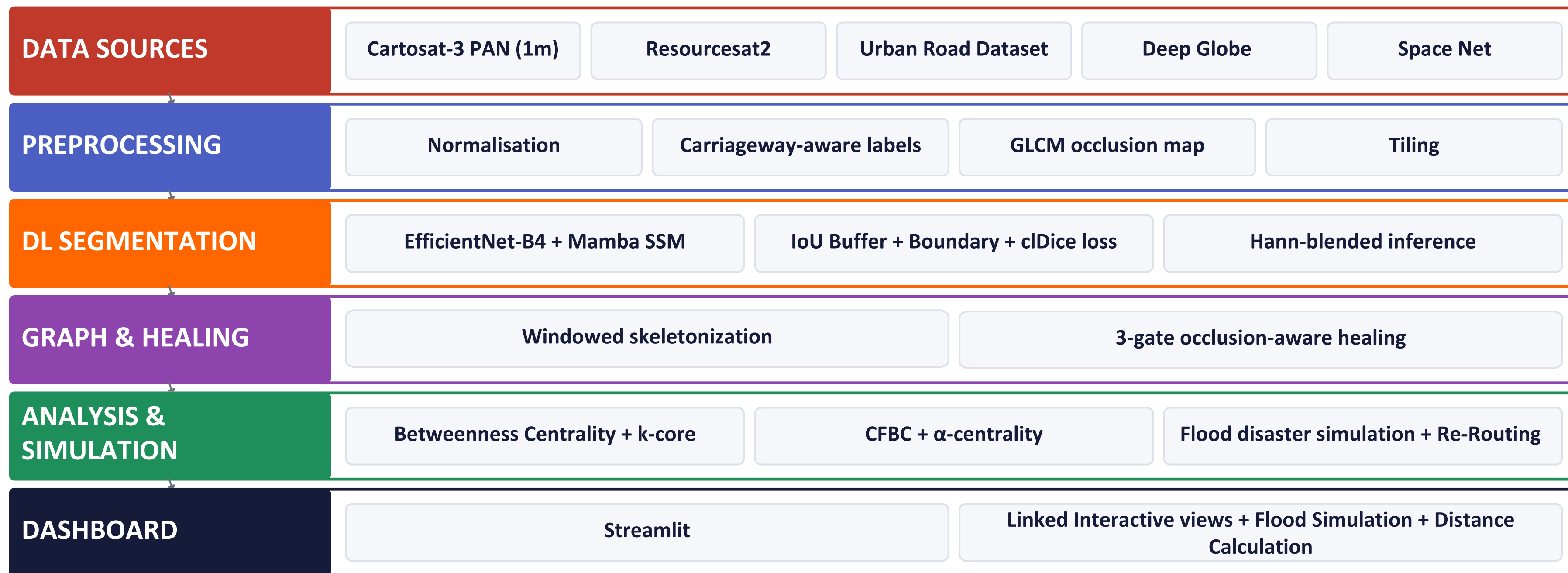


Technical approach



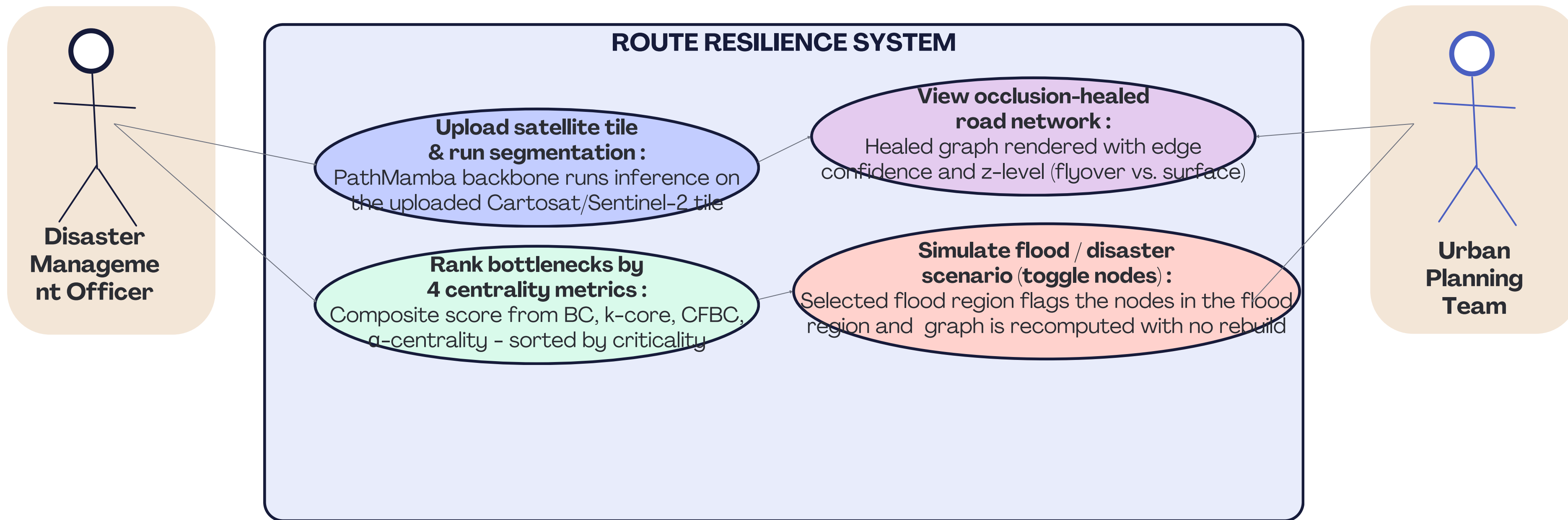
Architecture Diagram

Layered system architecture - from raw satellite tiles to a resilience-tested road graph.



Use-Case Diagram

How a disaster-response officer uses Route Resilience end to end, from upload to evacuation triage.



CORE FLOW Select flood zone → nodes flagged is_disabled → network recomputed



Results

53%

Buffered IoU — Cartosat-3 Fine-Tune



TRAINING LOG SNAPSHOT

```
[19:27:33] [C_cartosat] epoch 14 batch 350/374 loss=0.6352 ({'tversky': 0.45366042852401733, 'boundary': 0.04391828924417496, 'cldice': 0.26077669858932495, 'confidence': 0.23579540848731995, 'buffered_iou': 0.5332316160202026})
```

Results



Hyderabad Image

Results



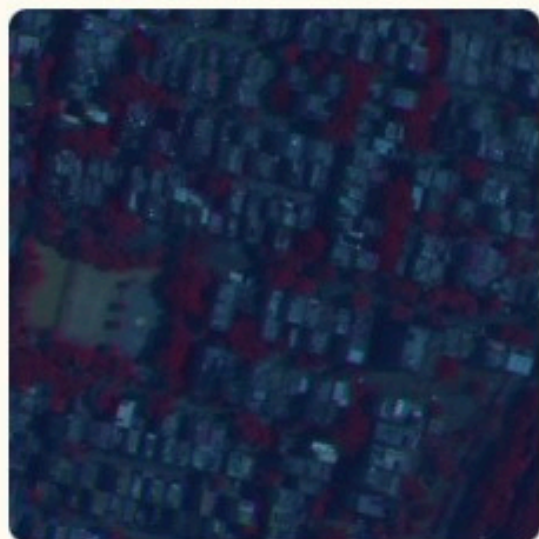
Hyderabad Segmentation

Other Results

Pipeline stages + network topology

Input tile → model's predicted road probability → final healed network → criticality heatmap, all in one row.

Input



Predicted (prob mask)



Healed network

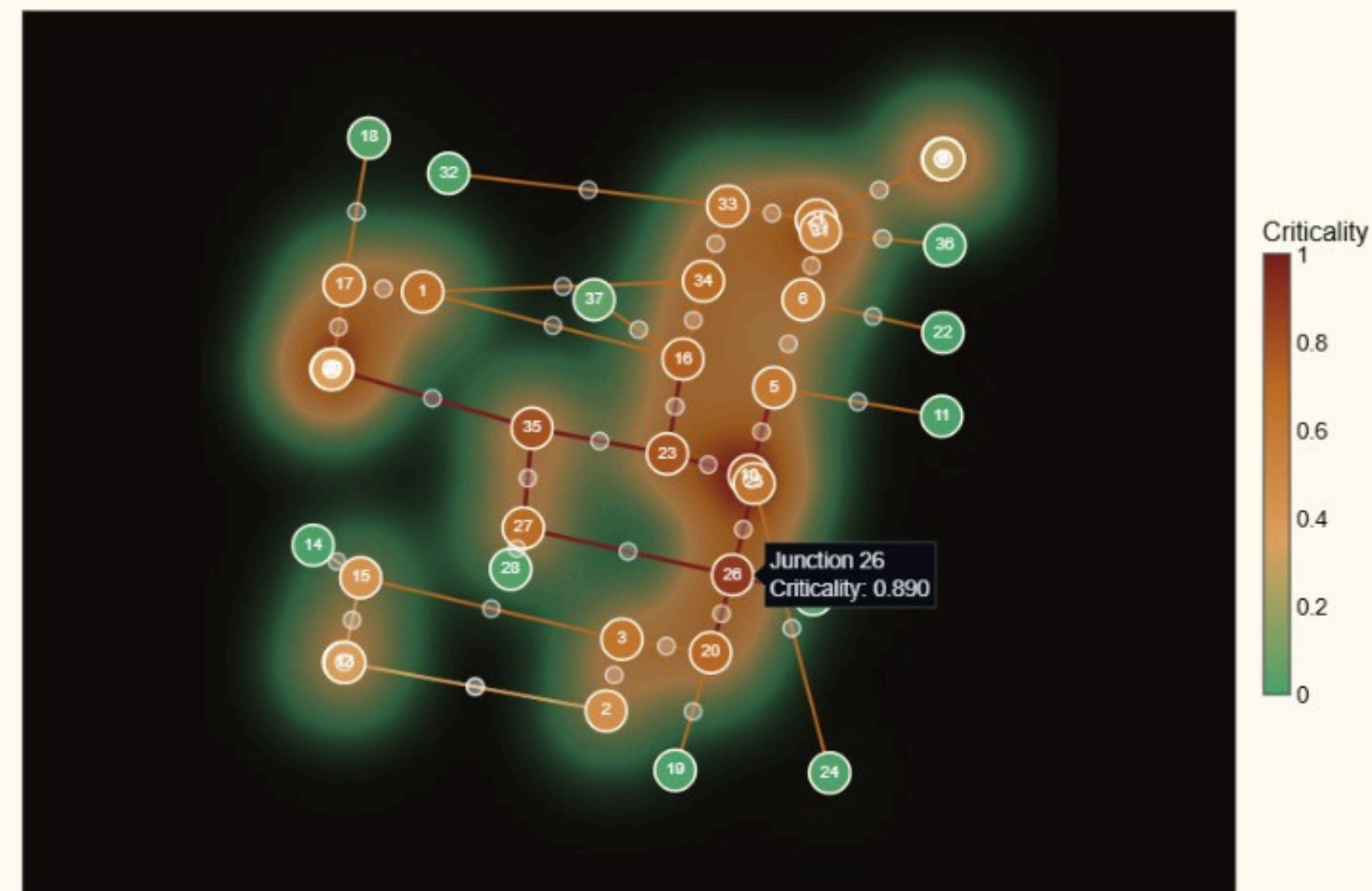


Network topology (criticality heatmap)

Colour by

- Composite (all 4 combined)
- Betweenness centrality
- K-core number
- Current-flow betweenness
- Alpha centrality

Darkers red = more critical junction or road segment (each metric normalized to 0-1; segment color = the higher of its two endpoints).



Simulate a flood

Flood origin junction

1

Steps



Randomize origin

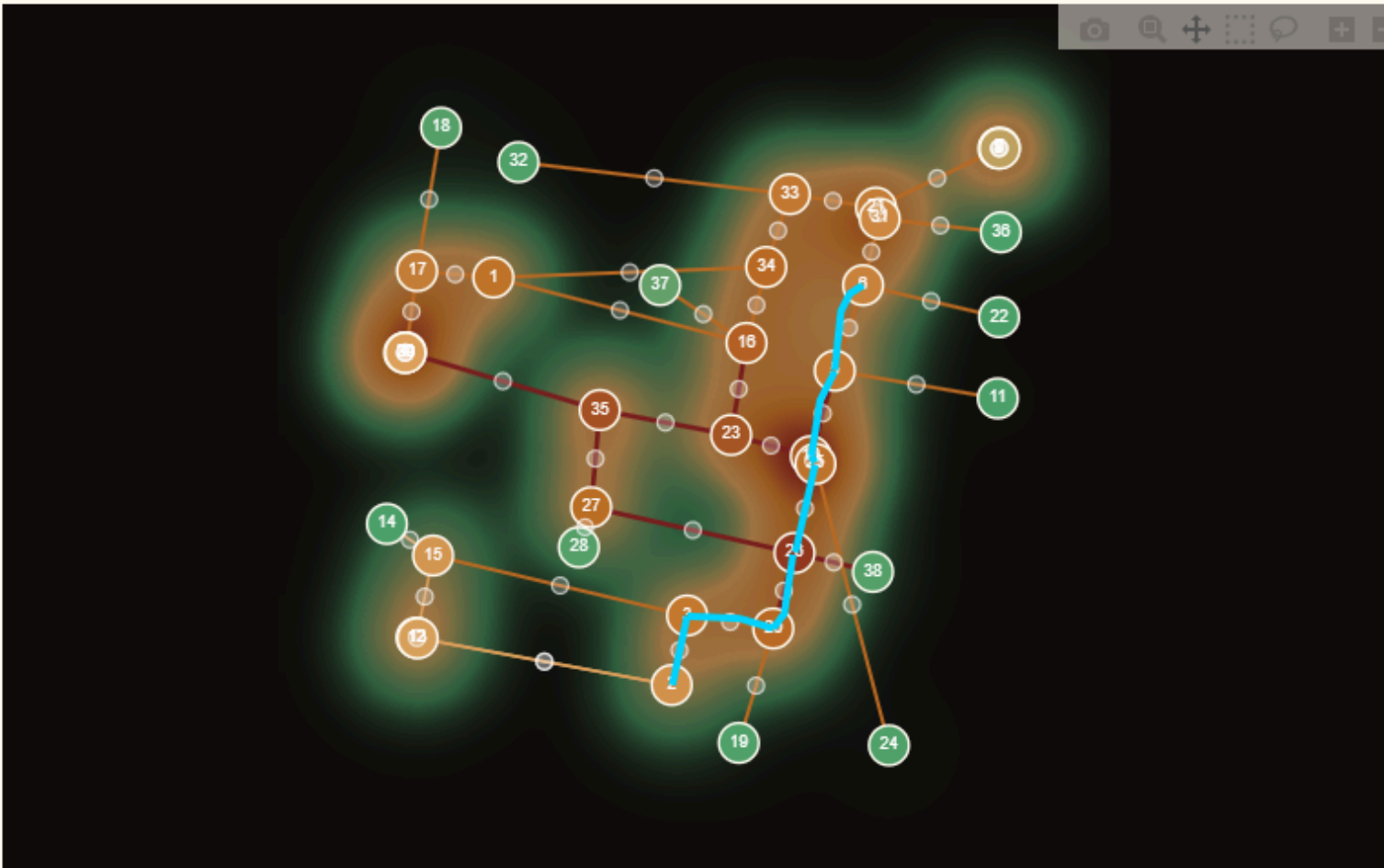
Play flood

Flood progress (step)

0



Click to select (before)



Result (after)

Click one or more junctions or roads on the left to simulate a disruption.

Selected route: baseline

Distance (no disruption)

Click to select (before)



Currently broken: junction(s) 5 and road(s) 10-25

Network intact

95%

Selected route: before vs after

Distance before

125 m

Distance after

263 m

Change

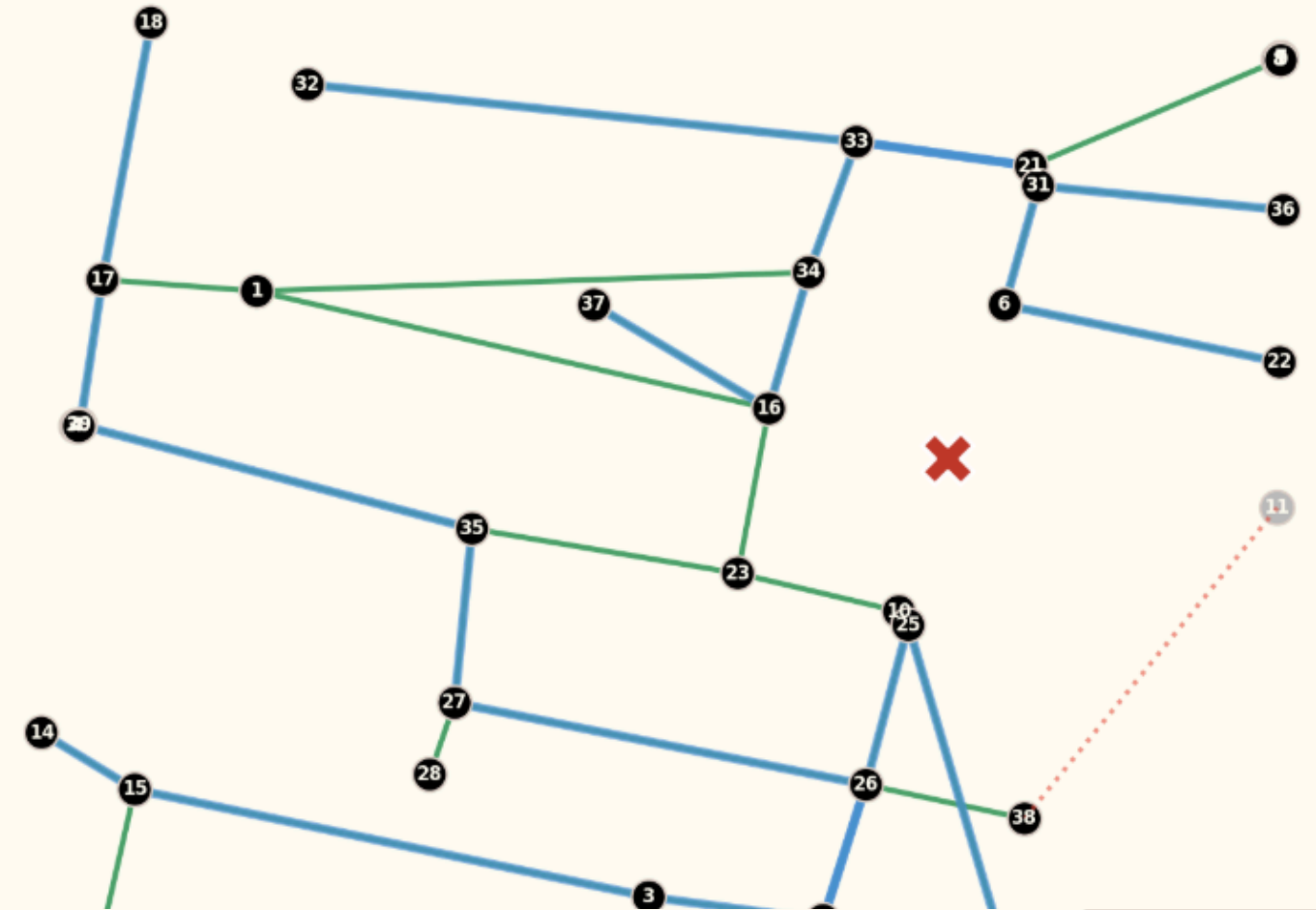
+111%

↑ +139 m

Result (after)

Visual reflects removed junctions only; the metrics below include both removed junctions AND removed road

After the break



Routes blocked

1 / 5

Flood origin junction

18

Steps

8

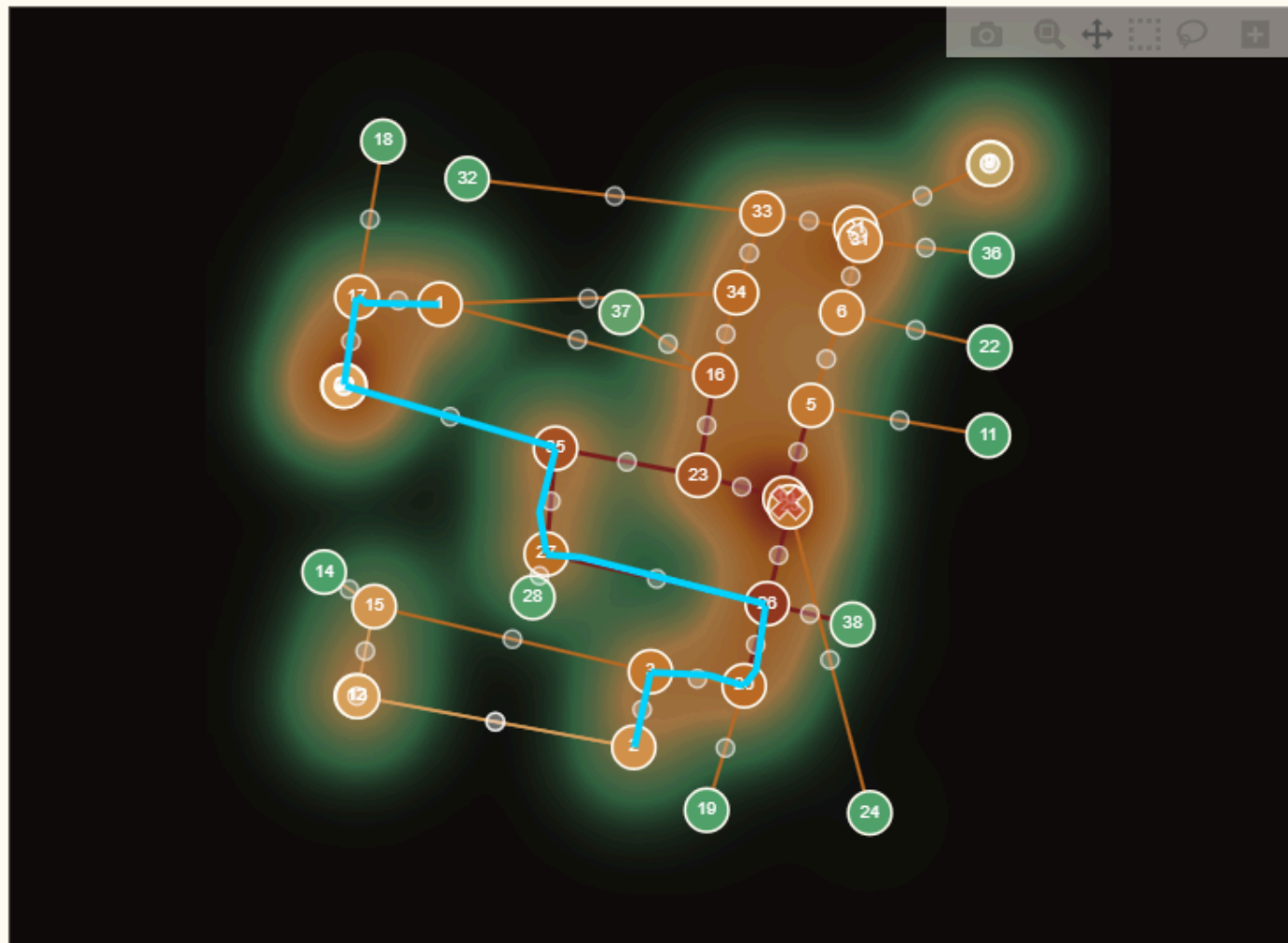
Randomize origin

Play flood

Flood progress (step)

0

Click to select (before)



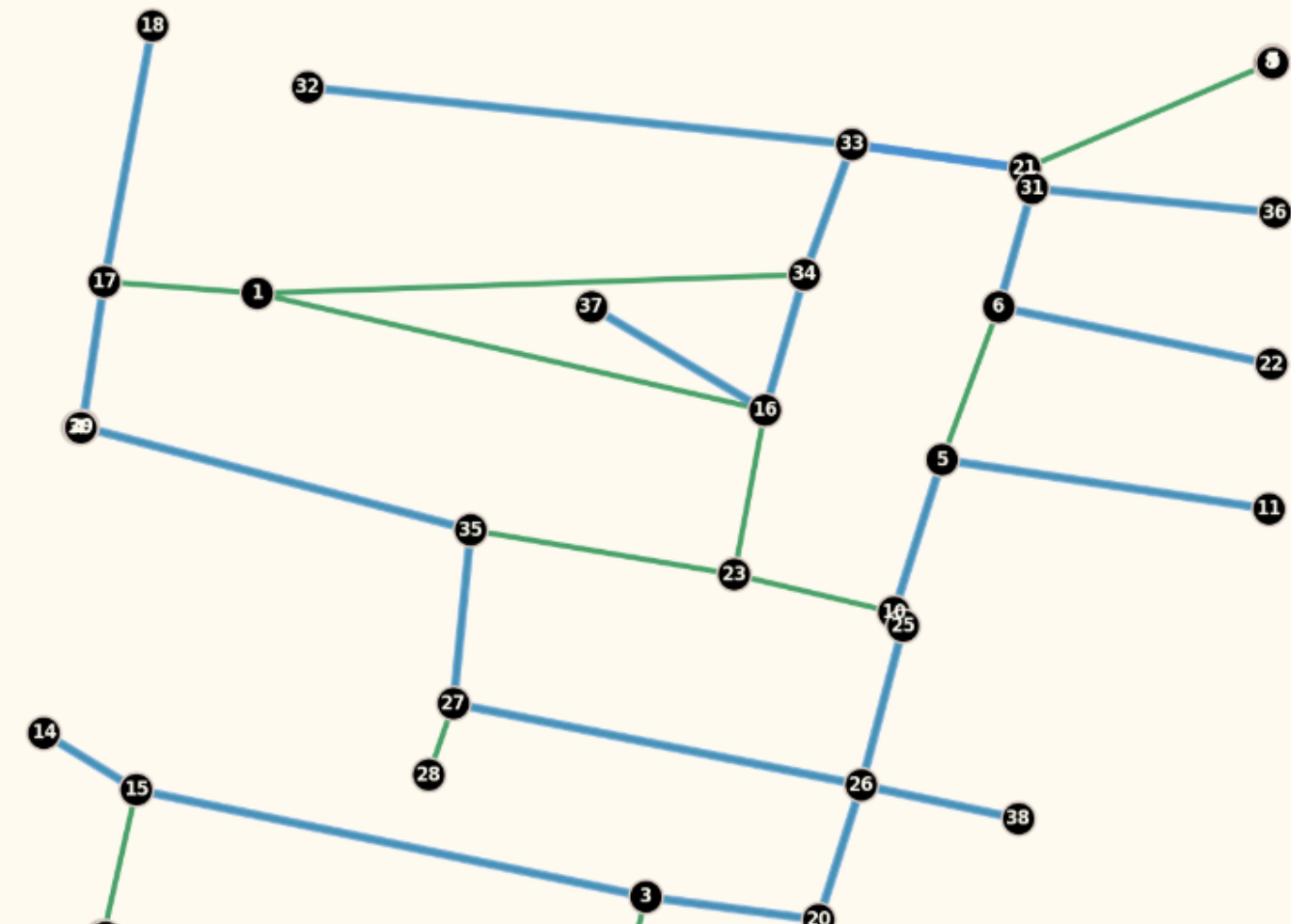
Currently broken: junction(s) 24 and road(s) 10-25

Clear selection

Result (after)

Visual reflects removed junctions only; the metrics below include both removed junctions AND removed roads.

After the break



18

8

Randomize Origin

Play flood

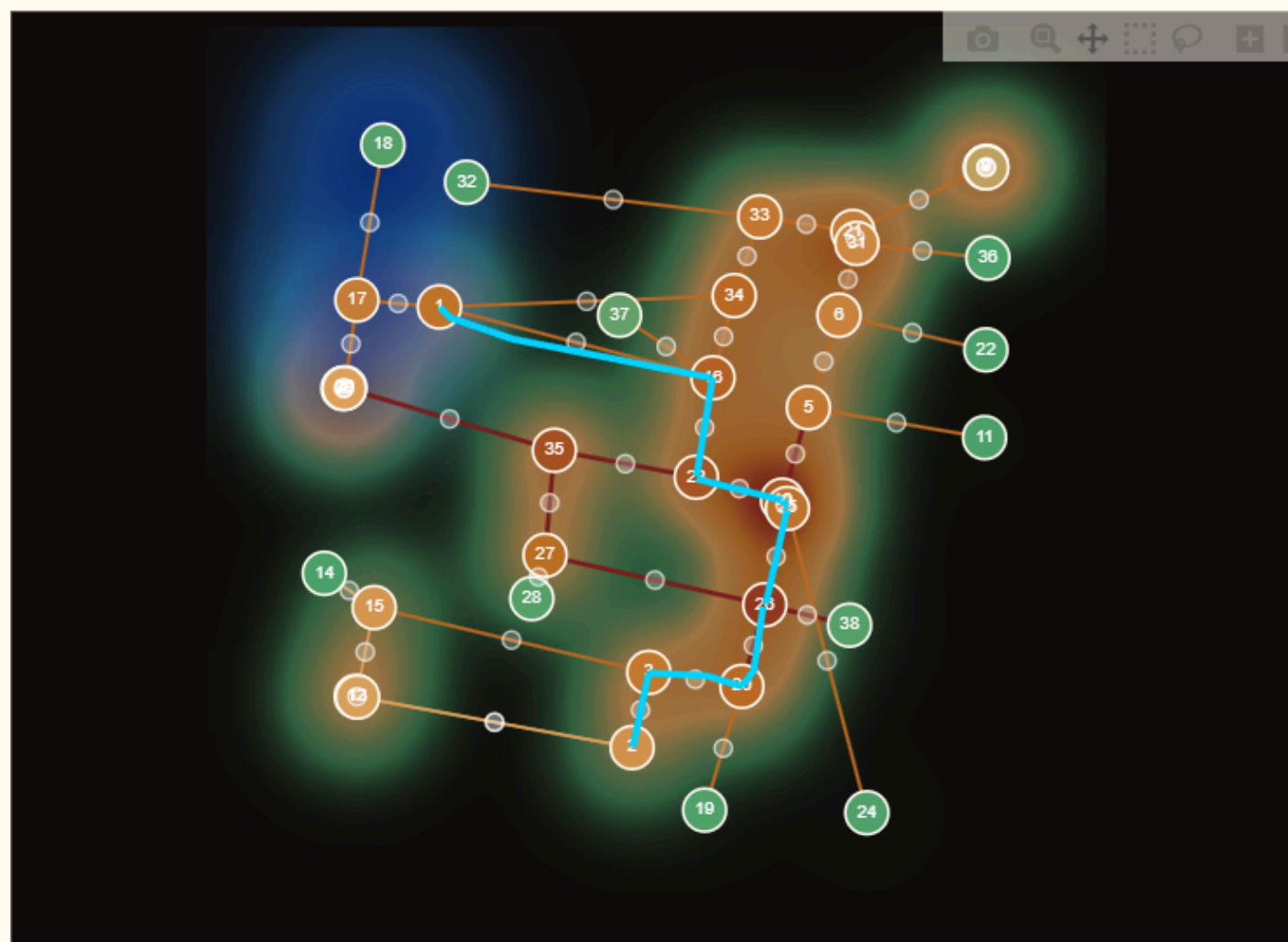
Flood progress (step)

1

Flood step 1/8 — 2 junction(s) underwater, spreading from junction 18.

Reset flood

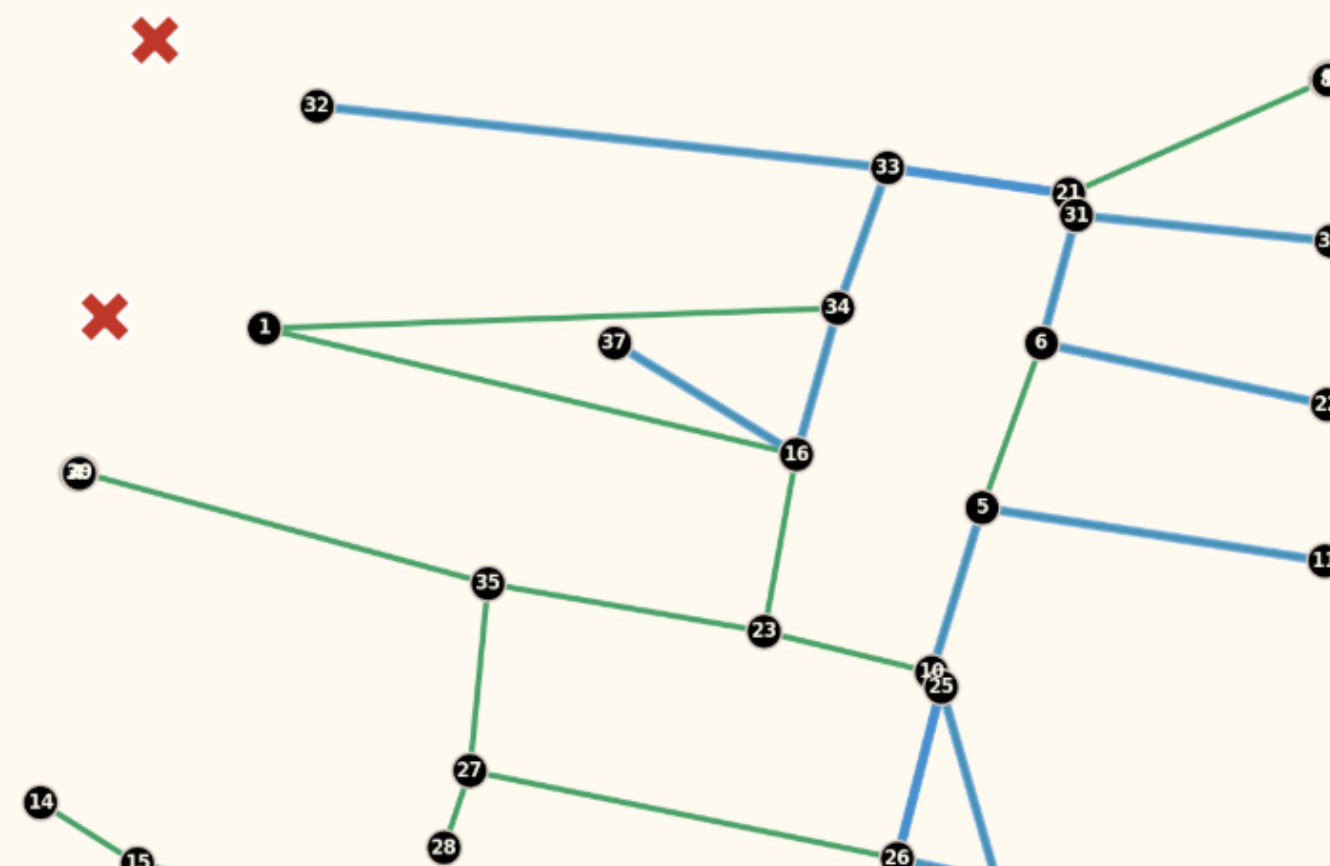
Click to select (before)

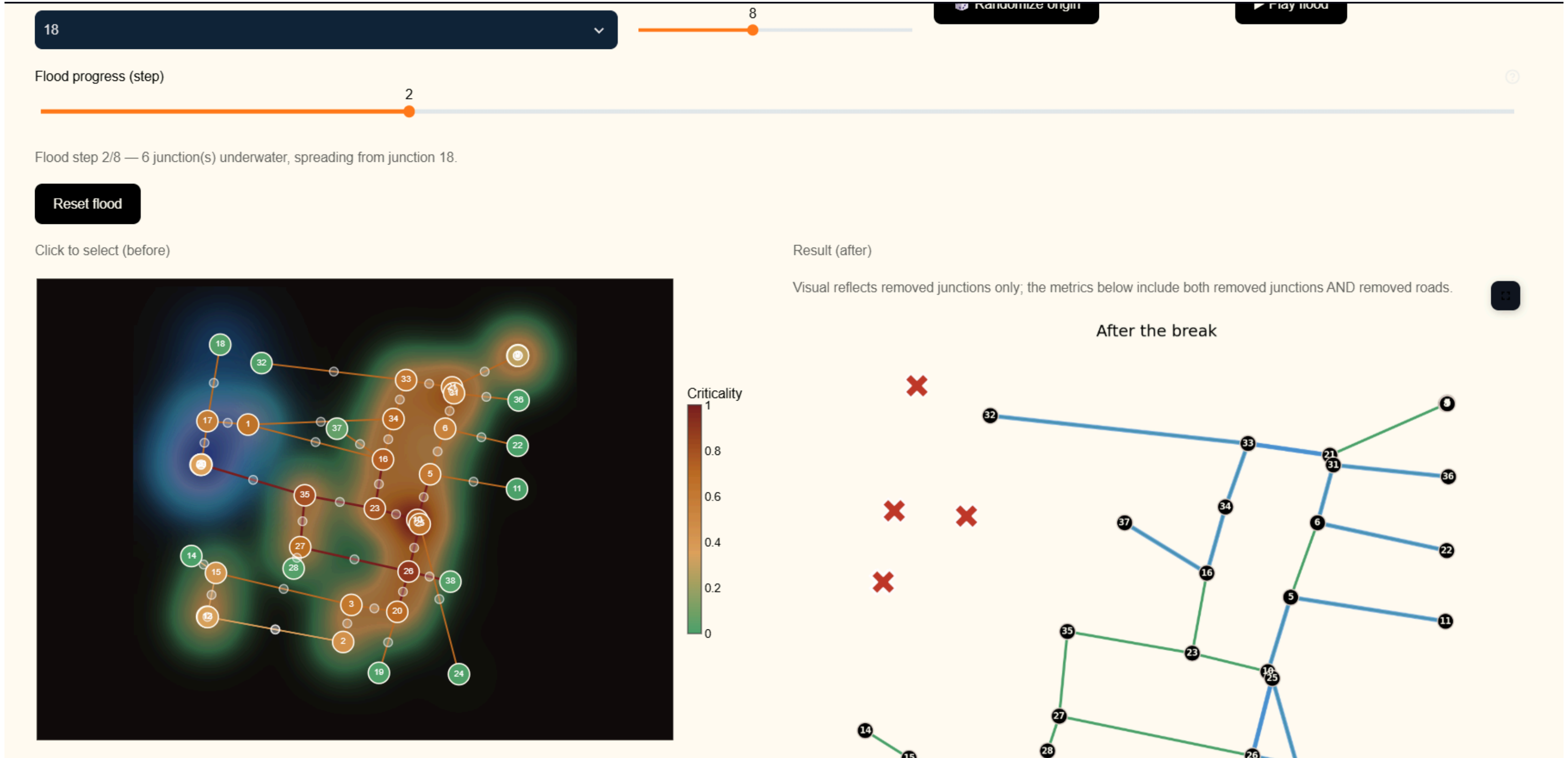


Result (after)

Visual reflects removed junctions only; the metrics below include both removed junctions AND removed roads.

After the break





1

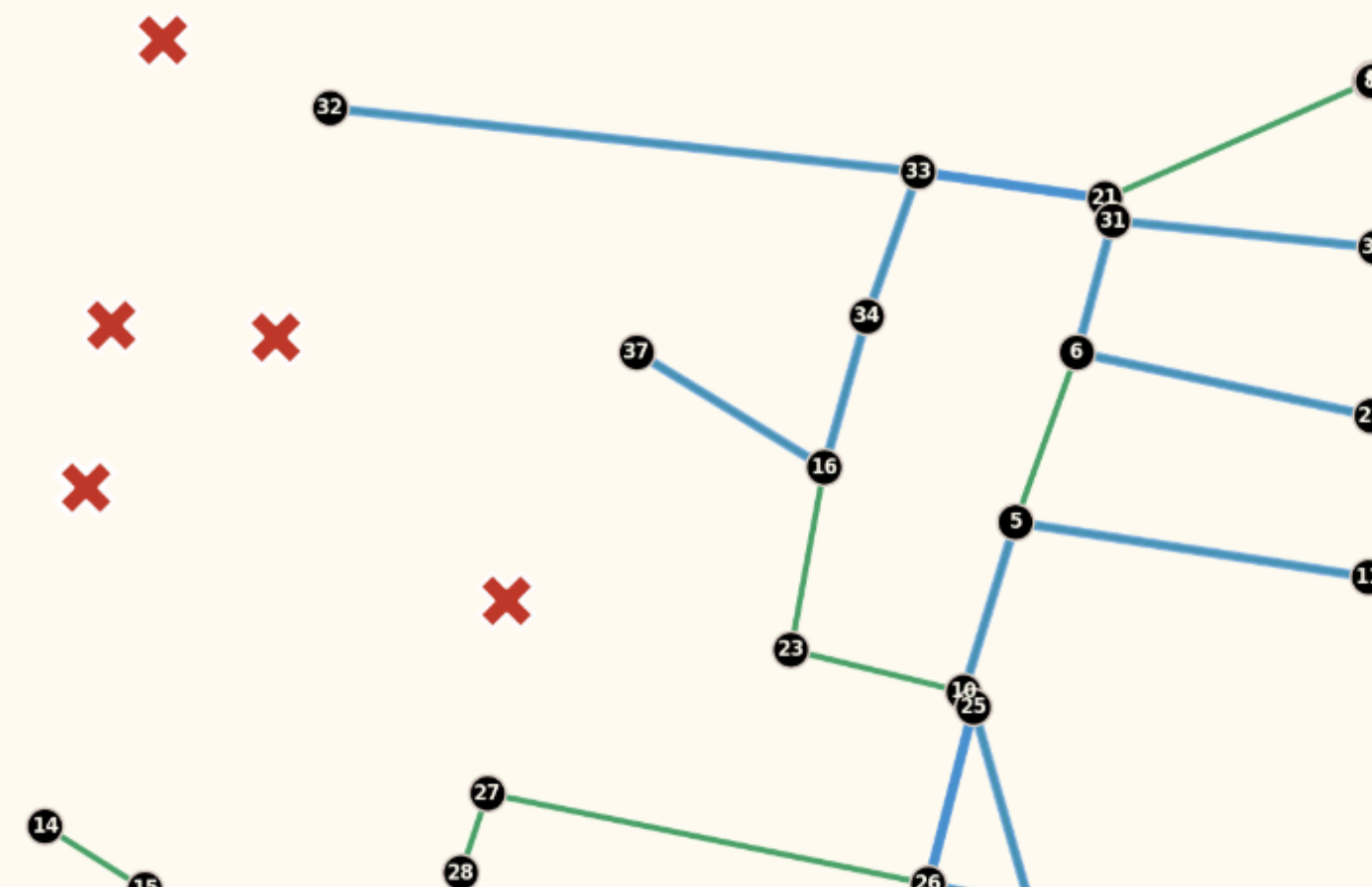
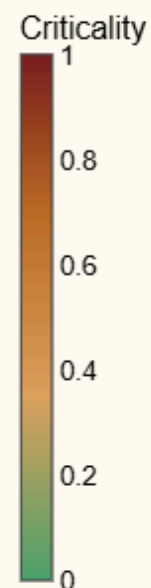
2

Reset flood

Result (after)

Visual reflects removed junctions only; the metrics below include both removed junctions AND removed roads.

After the break



18

8

4

Flood progress (step)

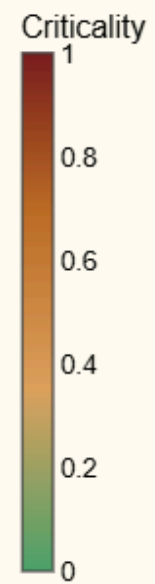
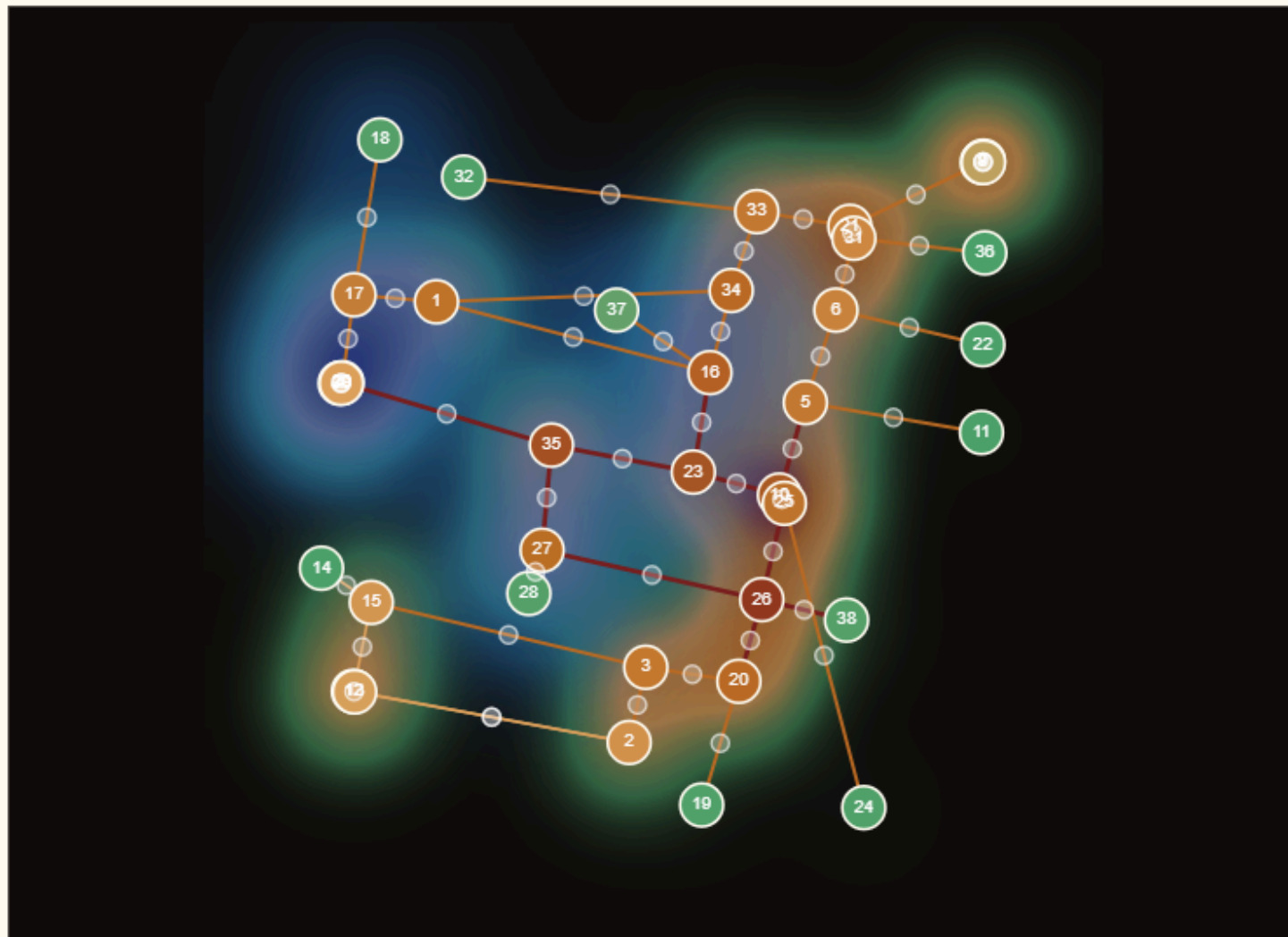
Flood step 4/8 — 13 junction(s) underwater, spreading from junction 18.

Reset flood

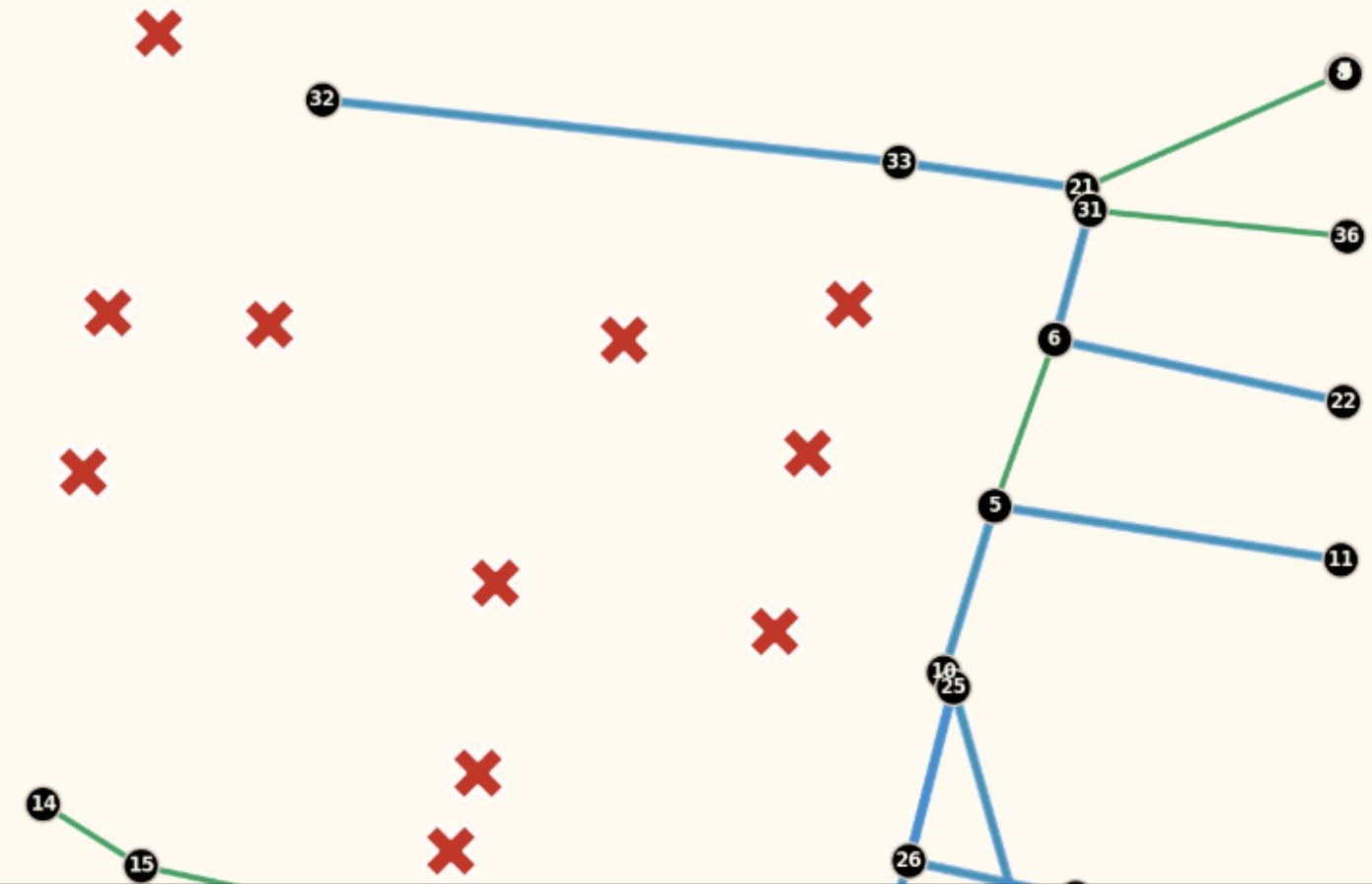
Click to select (before)

Result (after)

Visual reflects removed junctions only; the metrics below include both removed junctions AND removed roads.



After the break





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2026

THANK YOU